

UA Flight Operations QRH

OPS-O-004 Rev. 01

14-Sep-2026

The checklist will be completed by two crew members using a read-and-verify method. One crew member reads each checklist item aloud, and the second crew member verifies the action and confirms that the step has been completed correctly.



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0. Introduction

0.1 Foreword

This QRH (Quick Reference Handbook) document includes the checklists to be performed for the Pre-Flight, Flight, and Post-Flight stages of UA operations.

Its aim is to enable safe flight operations, and to prevent damage or injury.

0.2 Record of Amendments

#	Date	Description of change	Page	Rev.	Entered by	Signature
1	14-Sep-2026	Translation from Hebrew to English (FD QRH document)	All	01	Etay Vider	<i>Etay Vider</i>

0.3 Approvals

Action	Name	Title	Signature	Date
Checked by	Ziv Cohen	Chief Pilot	<i>Ziv Cohen</i>	14-Sep-2026
Approved by:	Rea Slonimsky	Flight Deck Manager	<i>Rea</i>	14-Sep-2026

0.4 Notations

Notation	Explanation
WARNING! 	An operating procedure, practice and so forth, which if not followed correctly, could result in personnel injury or loss of life.
NOTE	An operating procedure, condition and so forth, to which special attention should be paid.



1. Preliminary Preparations

NOTE

Perform this stage only at the start of the flight day. It is not required before every flight.

1.1 Regulatory Compliance and Approvals

1.	Verify that a valid flight authorization is valid for the operating site and scheduled operating period.
2.	Review NOTAMs and any relevant airspace restrictions.
3.	Confirm compliance with altitude, range, operating-hour and aircraft-type limitations.

WARNING!



No valid authorization - no flight.

1.2 Surveying the Area of Operations

Physically survey the area of operations, while verifying the following:

1.	Horizontal and vertical operating-area boundaries.
2.	Verify orientation and situational awareness using cardinal directions (North, South, East, and West).
3.	Primary take-off and landing site.
4.	Potential emergency landing sites.
5.	Hazards, including people, structures, power lines and vehicles.
6.	Safety vulnerabilities, including blind spots, prohibited flight areas, sensitive installations and other activities within the operating area.



1.3 Emergency Scenarios and Responses

Verify that the crew is familiar with, and ready to respond during the following emergencies:

1.	GNSS interference or disruption.
2.	Loss of communications (RC/telemetry).
3.	UA technical malfunction.
4.	Overheating, fire or battery swelling.
5.	A person entering the operating area.

NOTE

Verify safety equipment readily accessible, including a fire extinguisher and first-aid kit. Verify that MEL is complete.

NOTE

Before flights begin, review each emergency scenario, and plan for the corresponding response.

1.4 Weather

1.	Check current weather conditions and forecast.
2.	Confirm that conditions are within UA operating limitations.
3.	Wind conditions, including wind gusts.
4.	Temperature.
5.	Visibility.

WARNING!



If there is any doubt regarding weather, postpone flights.

2. Pre-Flight Checks

NOTE

Perform this stage only at the start of each operating session for the relevant UA. It does not need to be repeated between consecutive flights.

WARNING!



Verify that there are no open JIRA tickets for this UA.
A UA with an open JIRA ticket may be operated **ONLY** with the approval and attendance of an investigator.

2.1 Hardware Inspection – Structural Integrity

2.1.1 Propellers

1.	Correct orientation.
2.	Securely tightened.
3.	No cracks or deformation.
4.	Correct mounting for the UA type, such as motor washers or bolts tightened to the specified torque.

2.1.2 Fasteners

Make sure there are no loose fasteners.

2.1.3 Structural Integrity

1.	Perform a 360-degree visual inspection of the entire UA.
2.	Make sure there are no abnormalities such as impacts, dents, excessive dust or other damage

2.2 Software Checks

Ensure complete familiarity with the UA failsafe configurations and the expected automatic response for each emergency scenario:

1.	Loss of RC link.
2.	Loss of GNSS.
3.	Loss of GCS communications.
4.	Low battery.
5.	Critical battery.
6.	Vertical or horizontal geofence breach.

Ensure complete familiarity with:

1.	RC transmitter operation and all active switches and functions.
2.	Computer/GCS operation.



3. Pre Take-Off Checks

NOTE

Perform this stage before every flight.

3.1 RC Transmitter

1.	Visually inspect the RC transmitter.
2.	Check switches and control sticks for full, free movement and proper spring return.
3.	Verify that no components are loose and that there is no excessive mechanical play.

3.1.1 Before Power Up

Throttle	0
Manual flight mode	Manual/Stabilized
RTL	Disabled
Motor Stop	Enabled

3.1.2 After Power Up

1.	Verify that the transmitter is not bound to another UA.
2.	Verify that the RC is configured with the correct model.
3.	RC transmitter battery voltage is at least 7.4 V (<i>use only a two-cell battery</i>).
4.	Verify stick calibration using the transmitter display.
5.	Verify stick calibration using the operating software.



3.2 UA Inspection

Additional visual inspections:

1.	Correct propeller direction.
2.	Overall UA integrity.

3.2.1 Battery

1.	Battery is fully charged.
2.	Visually inspect the battery; confirm there is no swelling or damage and that connectors are in good condition (no carbonized connectors).
3.	Battery is balanced; cell-to-cell voltage difference does not exceed 0.1 V.
4.	Battery is physically secured.

3.2.2 Positioning

Check that the UA is correctly positioned on the take-off surface / in the docking station (where applicable).



3.3 Communications and Software Checks

1.	Connect (link) to the UA using the designated software.
2.	Verify that the site configuration is suitable for the area of operations.
3.	Verify there are no abnormal warnings.
4.	Verify valid telemetry.
5.	Verify the displayed nose/heading direction matches the UA orientation.
6.	Verify correct response to the RC flight-mode selections, and check response to RTL command.
7.	Disconnect or power off the RC transmitter and confirm that the FailSafe mode is activated.
	WARNING!  Before performing this check: 1. Verify how the failsafe performs (for example, does the UA have to be ON for the check). 2. Take precautions in case the failsafe is incorrectly configured – for example, propellers may activate.

On the first flight:

1.	Perform a radio calibration.
2.	Verify correct control-stick response in the software.
3.	Verify that the communication link is stable and consistent.

4. Flight / Take-Off Checks

NOTE

Before automated flights begin, perform a manual sanity flight. Verify stability, correct stick response, smooth flight with no vibration or abnormal noise.

- | | |
|----|---|
| 1. | Verify that the take-off area is clear of people and obstacles. |
| 2. | Announce "TAKE-OFF" clearly and loudly. |

4.1 First Take-Off

Slow climb

Controlled

Verify correct control-stick responses.

Brief throttle openings/load checks, preferably above a safety net.



5. Landing

5.1 Before Landing

1.	Verify that the landing area is clear of people and obstacles.
2.	Perform a controlled landing.

WARNING!



Do not let go of the RC transmitter until the UAV has finally landed and the motors have been shut down.

5.2 Approaching the UA

1.	Approach the UA only after the UA has finally landed and motors have stopped completely
2.	Throttle is at 0 .
3.	Motor Stop is Enabled .

5.3 Power-Down Sequence

1.	Power down the UA first.
2.	Then power off the RC transmitter.

6. Post-Flight Checks

1.	Perform a post-flight visual inspection of the entire UA.
2.	Check the battery for temperature and swelling.
3.	Record all faults and irregularities.



7. Essential Checks for Automated Flight

NOTE

The checks required before an automated flight necessarily include all preceding checklist stages, with particular emphasis on Preliminary Preparation, Pre-Flight Aircraft Checks and Pre-Take-Off.

7.1 Iron Drone

1.	Verify that the UA version matches the GCS version.
2.	Verify that the DTC transmitter Preset matches the UA.
3.	Verify that site_config.yaml matches the operating site.
4.	Verify time synchronization between the UA and the GCS.
5.	Verify valid communications with the MGG canister.
6.	Verify that the altitude difference between the target and interceptor is correct according to the simulated scenario.
7.	Verify estimated_target_width matches the expected target size (propeller to parallel propeller).
8.	Verify that escape_after_shot = 1 .
9.	Verify the RC transmitter BIND, and verify response to flight control commands, before take-off.
10.	Verify that the Take-Off Station area is clear of people and obstacles.



7.2 Optimus

1.	Verify that the correct parameter file is loaded on the UA (Optimus 1EX - VRS+FLY).
2.	Verify that the UA is calibrated to the DSA.
3.	Verify that DSA azimuth is correctly defined in HQ
4.	Verify User Synchronized Take-off.
5.	Before mission launch, Verify the DSA cameras are activated for monitoring the DSA robotic arm, and the arm is ready at the Stop Motion position.
6.	Verify that the RC transmitter is bound to the UA, and verify flight mode operation (change in RC, and verify on HQ)
7.	Verify that the station area is clear of people and obstacles.



8. Automated Flight RC Control Reference

NOTE

For the Optimus 1 EX RC control reference, see *Chapter 9 RC for Optimus 1 EX*.

WARNING!



Before operations, always verify which switch assignments, stick commands and expected responses are required, as per the approved UA configuration. If there is any doubt, check with your supervisor

Control	Position / Command	Result
Optimus X		
Optimus Parachute	Default / Up	Up
Optimus Parachute	Down	Disarm
Motor Stop	Up	Not locked
Motor Stop	Down	Locked
Optimus ELS	Up	Not active
Optimus ELS	Down	Active
Optimus flight modes	Up / Middle / Down	Stabilized / Loiter / Auto
Optimus arming	Stick command shown in source controller diagram	Arm
Optimus disarming	Stick command shown in source controller diagram	Disarm
Raider		
Raider Arm	Rotary knob - All left	Disarm
Raider Arm	Rotary knob - All right	Arm
Raider RTL	Up	Not active
Raider RTL	Down	Active
Raider flight modes	Up / Middle / Down	Stabilized/Stabilized/Guided



Control	Position / Command	Result
Target		
Target RTL	Up	Not active
Target RTL	Down	Active
Target flight modes	Up / Middle / Down	Stabilized / PosHold / Auto

Table 1: RC controls reference



8.1 Taranis RC Control Basic Functions

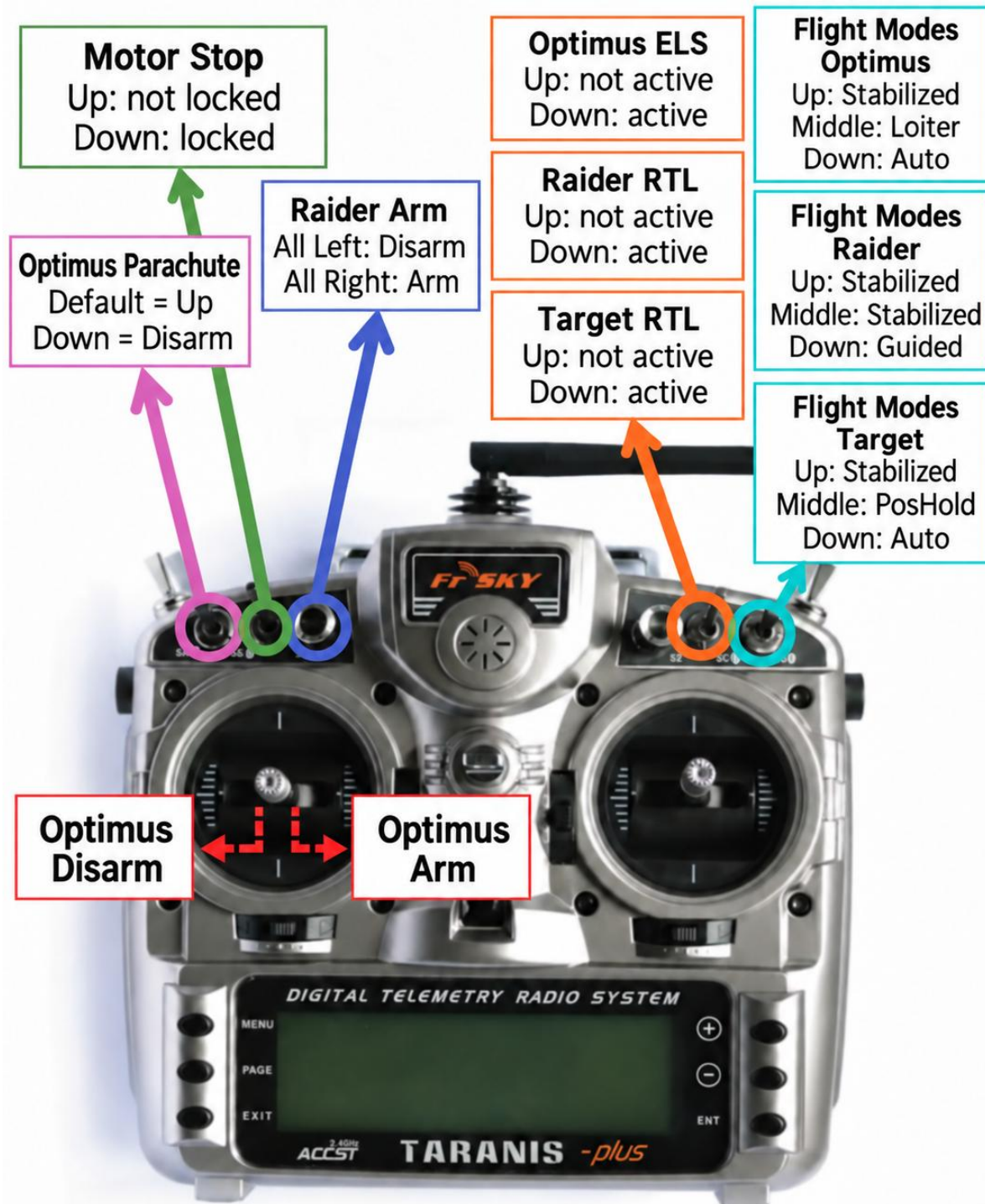


Figure 1: Taranis RC – basic functions

9. RC for Optimus 1 EX

The Optimus 1 EX system uses the Futaba RC transmitter for control.

9.1 Futaba RC Control Basic Functions



Figure 2: Futaba RC – basic functions

#	Control	Function	Action	Result
1.	Left Stick	Throttle	Push FORWARD Pull BACK	more thrust less thrust
		Yaw	Move LEFT Move RIGHT	Yaw LEFT Yaw RIGHT
2.	Right Stick	Pitch	Push FORWARD Pull BACK	Pitch DOWN Pitch UP
		Roll	Move LEFT Move RIGHT	Roll LEFT Roll RIGHT
3.	DLS switch	DLS	UP position	(Auto) Normal Flight
			DOWN position	Send to DLS
4.	Parachute	Parachute auto/disarm	UP position	Auto
			DOWN position	DISARM
5.	Flight mode switch	Select Flight mode	UP position	MANUAL mode
			CENTER position	POSITION mode
			DOWN position	AUTO mode

Table 2: Futaba RC controls reference

The RIGHT STICK is spring-loaded in all directions (forward/back and left/right) the stick will return to center when released.

The LEFT STICK is spring-loaded in the horizontal direction (left-right) – the stick will return to center when released. If the left stick is pulled back or pushed forward, it will remain in its vertical position – there is no spring loading vertically.



10. Terminology and Abbreviations

Term	Meaning
DLS	Designated Landing Site
DTC	Domo Tactical Communications (radio manufacturer)
ELS	Emergency Landing Site
Failsafe	Preconfigured aircraft response to a fault or loss of a required signal.
GCS	Ground Control Station.
Geofence	Software-defined operating boundary (horizontal and vertical)
GNSS	Global Navigation Satellite System
Guided	Navigate to single points guided by GCS
Loiter	Fight mode
MEL	Minimum Equipment List
NOTAM	Notice to Airmen.
PosHold	Position Hold flight mode.
QRH	Quick Reference Handbook
RC	Remote control / radio-control link.
RTL	Return-to-Launch – same as RH – Return Home
Stabilize	Self-levels roll and pitch axes: pilot manages throttle
UA	Unmanned Aircraft

